



3D printing filaments

3D printers come with a range of inks or filaments, here are their individual properties and use cases. <http://digit.in/sep20-5>



Rebuilding coral reefs

Marine scientists are using 3D printed terracotta tiles to encourage the growth of corals in Hong Kong. <http://digit.in/sep20-6>

3D PRINTING

An insight into the origins of this revolutionary modern-age technology

Dhriti Datta | dhriti@digit.in

U

nless you've been living under a rock, you'd have heard of the wonders of 3D printing. The seemingly nascent

technology may seem like it has been invented a few years ago, due to the extensive amount of media coverage and developments in the space, however, the technology was invented decades ago with ideas pertaining to it dating back even further. 3D printing, also known as additive manufacturing, is the process of constructing a three-dimensional object from a CAD (computer aided design) model or a digital 3D model. Holistically, 3D printing involves an array of processes where material is deposited, merged or solidified under computer control to

create a three-dimensional object. Materials such as liquid molecules or powder grains are added layer by layer and then are fused together. One of the primary uses of 3D printing is the ability to produce complex models that would be laborious to construct by hand. They are useful for prototyping products to see if it works fine before mass production can begin. Today, it is a widely accessible technology that can be purchased by anyone and used not only professionally, but also for recreation. So, with that being said, let's explore the history of 3D printing or additive manufacturing.

THE EARLY BREAKTHROUGHS

Some arguments can be made for processes such as sculpting to be the original forms of 3D printing since they transformed the two-dimensional world artists and product designers lived in, into a three-dimensional one. However, these sculpted models, also known as prototypes, are laboriously slow to build and can also be an expensive affair. While 3D printing as a technology wasn't born until the late twentieth century, some ideas pertaining to the technology came to life much earlier. In

1859, French photosculptor, François Willème, demonstrated the world's very first '3D scanning or 3D modelling' technology named sculpture photographique (photographing sculpture) where 24 cameras (placed at every 15-degrees) were simultaneously used to photograph subjects from different angles to build an accurate 3D modelled sculpture. In 1892, the inventor, Joseph E. Blather, was awarded a patent for a method of creating 3D topographical maps by utilising a layering method which is similar to the concept used in today's 3D printers. These ideas and inventions possibly paved the path for the breakthrough technology that is 3D printing.

THE BIRTH OF 3D PRINTING

The earliest forms of 3D printing technology surfaced in the 1980s. They were termed as Rapid Prototyping (RP) technologies since they were meant to be a faster and more cost-efficient method of creating prototypes for product development. In the year 1980, when most people were marvelling at the capabilities of a regular printer, Dr Hideo Kodama of Nagoya Municipal Industrial Research Institute first invented an additive method for fabricating 3D plastic models with photopolymers. In November 1981, Kodama filed for a patent for his rapid prototyping system. Unfortunately, the Japanese doctor failed to follow up on his patent application and wasn't able to commercialise his printing process.

Nevertheless, the idea of 'rapid prototyping devices' continued to steamroll ahead and in 1984, Chuck Hull invented a process known as stereolithography that allows designers to create tangible 3D models using digital data. In this method, objects are printed layer by layer using a process in which UV lasers selectively cause chains of molecules (in liquid resin) to link to one another, forming what are known as polymers. The technology created quite a lot of stir among inventors, since they could now theoretically prototype and test designs without having to make enormous upfront investments in manufacturing. Hull applied for a patent for his technology which was issued in 1986, and in the